Cat Spacer

Short manual . free and open source (AGPL-3.0-or-later) . github.com/SpeakingColors

Cat Spacer is a spectral sidechain ducker. Instead of ducking the whole sound, it listens to the sidechain (for example a vocal or kick) and cuts only the frequencies in your main track where the sidechain is loud, so both can sit in the same mix. No threshold is needed: it follows the loudest sidechain bands.

Quick start

• Put Cat Spacer on the sound you want to carve (a lead or pad) and route the sound that should stay clear (vocal, kick) to its sidechain input in your DAW.
• Raise Amount until the gaps appear in the display; the cut curve shows what is being removed. At 0 nothing is ducked; low settings carve only the bands where the sidechain is loud; towards 100 it becomes a full pump that ducks everything on every hit.
• Use Low cut and High cut to limit the range that is affected, and Max cut to set the deepest cut.
• Use Audition to hear the sidechain as the plugin hears it, Freeze to hold the current sidechain shape.
• Drive, Transistor and Fuzz (PurrOne colour stages, up to 150 percent) dirty the sound; the Colour menu chooses where: sound (level kept automatically), sidechain (what the ducker listens to), both, or not ducked (only the part below Low cut and above High cut, which the sidechain never touches).
• The round gear button in the header opens Settings: the plugin version, a link to the GitHub page and the releases page (it opens your browser; the plugin itself never goes online), and a switch for the info bubbles. Hover any control for about one second to see what it does; the round i button also turns the bubbles on or off.

How the window works

• Each knob sits directly under the display it changes; the displays react to the knobs and the knobs react to the displays.
• Meters show the real signal in dB with a dB scale: input and output on the far right.
• Hover a control for about one second to see an info bubble. Turn bubbles off with the round i button in the header.
• The Limiter switch in the header adds a safety limiter at the output (useful while experimenting).
• Presets are in the header; the dice picks random settings (some controls are excluded on purpose).

Controls

Spectrum and cut

Colour	Choice: Colour: sound, Colour: sidechain, Colour: both, Colour: not ducked (default Colour: sound)	Where Drive, Transistor and Fuzz act. Sound: dirties the audio you hear, level kept. Sidechain: only colours what the ducker listens to (Audition shows it). Both: does both. Not ducked: colours only what the sidechain never touches, below the Low cut and above the High cut (needs those set inside 20 Hz to 20 kHz).
Freeze	Choice: Off / On (default Off)	Freeze: holds the current sidechain curve so it keeps carving the same gaps (a static EQ taken from the sidechain).
Audition	Choice: Off / On (default Off)	Audition: listen to the sidechain as the plugin hears it (after low and high cut). Useful for setting the cuts.
Amount	0 to 100 % default 50 %	Amount: 0 = no ducking at all. Low settings carve only the bands where the sidechain is loud (transparent space). Towards 100 it turns into a full-on pump that ducks everything with each hit. Gradual in between.
Low cut	20 to 20000 Hz default 20 Hz	Low cut: sidechain content below this frequency does not trigger any ducking (stops a kick from carving the whole mix, for example).
High cut	20 to 20000 Hz default 20000 Hz	High cut: sidechain content above this frequency does not trigger any ducking.
Max cut	3 to 36 dB default 24 dB	Max cut: the deepest reduction any band can get at 100 % amount.
Floor	10 to 70 dB default 50 dB	Floor: how far below the loudest sidechain band a band can be and still cause ducking. Lower = only the strongest bands duck, higher = more bands duck.
Drive	0 to 150 % default 0 %	Drive (PurrOne opamp tanh). Colour: sound dirties the audio you hear with automatic gain staging so it never gets quieter. Colour: sidechain colours what the ducker listens to: more harmonics means more bands duck, so the cut gets wider.
Transistor	0 to 150 % default 0 %	Transistor (PurrOne, asymmetric, even harmonics). On the sound: warm, thick even harmonics with the level kept. On the sidechain: the ducker also reacts one octave up from the fundamental.
Fuzz	0 to 150 % default 0 %	Fuzz (PurrOne lopsided clip with octave-up growl). On the sound: nasty fuzz with the level kept. On the sidechain: spreads the sidechain energy across the spectrum, so a bass or kick carves holes far up (and the result gets quieter).

Timing

Latency	Choice: Ultra 256, Low 512, Normal 1024, High 2048 (default Low 512)	Latency: the delay this plugin adds (the host compensates). Ultra 256 is about 6 ms, Low 512 about 11 ms, Normal 1024 about 21 ms, High 2048 about 43 ms. Shorter = tighter timing but less precise at low frequencies; longer = cleaner and more exact in the bass.
Attack	1 to 300 ms default 20 ms	Attack: how fast the ducking engages when the sidechain gets loud.
Release	10 to 2000 ms default 160 ms	Release: how long the ducking takes to let go after the sidechain stops. Short = tight, long = the big breathing pump.

Stereo

Stereo mode	Choice: L/R / M/S (default L/R)	Stereo mode: L/R ducks left and right, M/S ducks mid and side.
Pan	-100 to 100 default 0	Pan: moves the effect toward one side. Left of centre applies less to the right channel, right of centre applies less to the left (or to side instead of mid in M/S).
Mix	0 to 100 % default 100 %	Mix: dry/wet blend of the ducked signal.

Recommendations

• Lead under a vocal: Amount 50 to 70, Attack 5 to 20 ms, Release 100 to 200 ms, Low cut 150 Hz, High cut 8 kHz.
• Pad under a kick: Release shorter (60 to 120 ms), Max cut 8 to 12 dB.
• Keep Floor above 0 if the carving sounds hollow; lower Mix to blend with the untouched sound.
• Pumping: Amount 100, Attack 5 ms, Release 200 ms, then shorten or lengthen Release to the tempo.
• Latency: Ultra 256, Low 512, Normal 1024, High 2048 samples plus a few samples for the colour stage; the host compensates. Use Low while playing live, High for the cleanest cut when mixing.

Interface tour

Pink numbers mark controls, teal letters mark displays. Every number is explained in the tables on the next pages. The picture is a screenshot of the real compiled plugin window.



Signal flow

Main input and sidechain input > 64 band spectrum analysis > cut curve > applied to the main signal (and mixed with a broadband pump as Amount rises) > optional colour stage (Drive, Transistor, Fuzz) > Mix, Pan, M/S > output.

Header bar

1	**Preset menu**	Factory presets (our own). Use the arrows to step through them.
2	**Randomize**	Set the controls to random values (output level and on/off switches are kept).
3	**Nasty range**	Nasty range: lets Drive, Fuzz, Squeeze, Warm and the other heavy knobs go far past 100 % for really nasty, OTT and Vengeance style sounds. Off keeps every knob at its normal top.
4	**Limiter**	Safety limiter on the output: keeps the level from ever passing 0 dBFS, smooth and transparent. Turn it off if you want the raw output (for example to let your own limiter do the work).
5	**Settings**	Settings: version, link to the GitHub page for new versions
6	**Info bubbles (i)**	Info bubbles: hover a control for one second to see what it does. Click to turn them off or on.

A. Spectrum and cut



Display A: Main display, frequency on the horizontal axis (log, 20 Hz to 20 kHz), dB on the vertical axis. Grey filled shape: the main signal spectrum. Blue line: the sidechain spectrum as the plugin hears it (coloured and after the cuts). White line: the cut applied in dB below 0, which is what is being carved out of the main sound right now. Two orange vertical lines mark Low cut and High cut: the sidechain is ignored outside them.

7	**Amount**	Amount: 0 = no ducking at all. Low settings carve only the bands where the sidechain is loud (transparent space). Towards 100 it turns into a full-on pump that ducks everything with each hit. Gradual in between. (shown: 50 %)
8	**Low cut**	Low cut: sidechain content below this frequency does not trigger any ducking (stops a kick from carving the whole mix, for example). (shown: 20 Hz)
9	**High cut**	High cut: sidechain content above this frequency does not trigger any ducking. (shown: 20000 Hz)
10	**Max cut**	Max cut: the deepest reduction any band can get at 100 % amount. (shown: 24.0 dB)
11	**Floor**	Floor: how far below the loudest sidechain band a band can be and still cause ducking. Lower = only the strongest bands duck, higher = more bands duck. (shown: 50.0 dB)
12	**Drive**	Drive (PurrOne opamp tanh). Colour: sound dirties the audio you hear with automatic gain staging so it never gets quieter. Colour: sidechain colours what the ducker listens to: more harmonics means more bands duck, so the cut gets wider. (shown: 0 %)
13	**Transistor**	Transistor (PurrOne, asymmetric, even harmonics). On the sound: warm, thick even harmonics with the level kept. On the sidechain: the ducker also reacts one octave up from the fundamental. (shown: 0 %)
14	**Fuzz**	Fuzz (PurrOne lopsided clip with octave-up growl). On the sound: nasty fuzz with the level kept. On the sidechain: spreads the sidechain energy across the spectrum, so a bass or kick carves holes far up (and the result gets quieter). (shown: 0 %)
15	**Colour**	Where Drive, Transistor and Fuzz act. Sound: dirties the audio you hear, level kept. Sidechain: only colours what the ducker listens to (Audition shows it). Both: does both. Not ducked: colours only what the sidechain never touches, below the Low cut and above the High cut (needs those set inside 20 Hz to 20 kHz). (shown: Colour: sound)
16	**Freeze**	Freeze: holds the current sidechain curve so it keeps carving the same gaps (a static EQ taken from the sidechain).
17	**Audition**	Audition: listen to the sidechain as the plugin hears it (after low and high cut). Useful for setting the cuts.

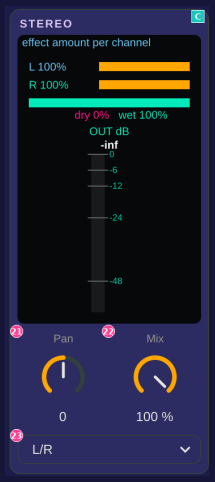
B. Timing



Display B: Shape of one ducking hit: depth against time. The curve falls at the Attack time and recovers at the Release time; both are printed under the graph.

18	**Attack**	Attack: how fast the ducking engages when the sidechain gets loud. (shown: 20 ms)
19	**Release**	Release: how long the ducking takes to let go after the sidechain stops. Short = tight, long = the big breathing pump. (shown: 160 ms)
20	**Latency**	Latency: the delay this plugin adds (the host compensates). Ultra 256 is about 6 ms, Low 512 about 11 ms, Normal 1024 about 21 ms, High 2048 about 43 ms. Shorter = tighter timing but less precise at low frequencies; longer = cleaner and more exact in the bass. (shown: Low 512)

C. Stereo



Display C: Per channel amount bars (L and R), a dry/wet bar (pink dry, teal wet) and an output level meter in dB.

21	**Pan**	Pan: moves the effect toward one side. Left of centre applies less to the right channel, right of centre applies less to the left (or to side instead of mid in M/S). (shown: 0)
22	**Mix**	Mix: dry/wet blend of the ducked signal. (shown: 100 %)
23	**Stereo mode**	Stereo mode: L/R ducks left and right, M/S ducks mid and side. (shown: L/R)

Under the hood

• Amount is a true blend: at 0 % the plugin is a bit-exact bypass (tested to -150 dB); low values carve only the bands where the sidechain is loud; towards 100 % the carve turns into a full broadband pump that ducks everything on every hit. Measured after a kick: about -3, -7, -12 and -19 dB at 25, 50, 75 and 100 %.
• No threshold: a band is cut when it is within Floor dB of the loudest sidechain band, down to at most Max cut dB.
• Colour menu: Sound colours the whole sound before ducking (level kept automatically); Sidechain colours only what the ducker listens to (it widens the cut and costs about 3 LU of loudness); Both does both; Not ducked colours only the part below Low cut and above High cut, which the sidechain never touches.
• Drive, Transistor and Fuzz go to 150 % (300 % for the nastiest settings) and come from PurrOne.
• Latency: Ultra 256, Low 512, Normal 1024, High 2048 samples (plus a few for colour). The plugin reports it exactly and the host compensates, so tracks stay aligned.
• Freeze holds the current sidechain curve; Audition lets you hear the sidechain exactly as the plugin hears it.

Install and use

• Windows: run the installer (.exe). macOS: open the installer (.pkg or .dmg). Linux: install the Flatpak with `flatpak install --user` and the .flatpak file, or copy the .vst3 folder to ~/.vst3.
• Everything works offline. Rescan plugins in your DAW, then look for the plugin under Studio 87.
• Report problems or ideas on the GitHub page of the plugin. Free to use, copy and change under the AGPL-3.0-or-later licence.